

Payment & Financial Services - Document 18 - Page 1/6

Document ID: PAY-018 | Classification: Internal Use

Financial reporting: transaction reports, revenue analytics, and regulatory reporting formats.

Multi-currency: store amount in minor units (cents). Display in major units. Use ISO 4217 codes

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Duplicate transaction prevention: Use idempotency keys. Check idempotency_key in request headers.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Payment SLA and uptime: 99.9% target, incident response times, and status page communication.

Fraud score > 80: block transaction. Score 50-80: require 3DS. Score < 50: auto-approve.

Webhook delivery failures: Verify endpoint URL, SSL certificate, and signature validation logic

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Error code 5003: Payment gateway connection timeout. Check network connectivity and gateway health.

Audit log retention: 7 years for financial transactions. Immutable storage. Tamper-evident.

Payment gateway integration architecture and secure token handling procedures for PCI-DSS compliance.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

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Error code 4001: Invalid payment method. Verify card details and expiry. Support 3DS if required.

Incident severity: P1 payment down, P2 degraded, P3 single merchant. P1 requires 15-min acknowledgment.

Transaction processing workflows and error recovery mechanisms for financial services including retry logic and idempotency.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

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Error code 4002: Insufficient funds. Customer must use alternative payment method or add funds.

Payment gateway requires secure token handling. Never store raw card numbers. Use payment tokens.

Payment API integration guidelines: REST endpoints, authentication, webhook callbacks, and error code handling.

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Troubleshooting:

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EU payment failures: Verify PSD2 compliance, SCA completion, and issuer 3DS configuration.

EU region mandates PSD2 and Strong Customer Authentication (SCA) for card payments.

Financial transaction security protocols: TLS 1.3, tokenization, encryption at rest and in transit.

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